Questions for the day

- How can *we* develop efficient algorithms like MergeSort and QuickSort?
- What are some challenges and shortcomings of this approach?

Introduction to algorithm design

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Lecture 7

Outline

- Algorithm design strategies
  - Brute force
  - Divide-and-conquer
  - Dynamic programming

Algorithm design strategies

- "Templates" for creating algorithms
- Need to be adapted to specific problem
- Give you more tools for approaching difficult problems

- **Strategy 0:** brute force
  - A.k.a., exhaustive search
  - Test all possibilities for the solution
  - Report the correct/best solution

Brute force example

- Brute force sorting

Input: $data$: array to be sorted
Input: n: length of $data$
Output: permutation of $data$ such that
$data[1] \leq \ldots \leq data[n]$

```
Algorithm: BruteSort
repeat
    Find the next permutation of data to test
    if data is sorted then
        return data
    end
until no more permutations of data are left
error data cannot be sorted
```

Analysis: brute force

Pros

- Applicable to most problems
- *Always* gets the correct/optimum answer
- Easy to design and describe

Cons

- Almost always slowest solution
- Often infeasible
- Exponential or factorial number of tests are impractical for most realistic problems

Divide-and-conquer

- **Goal**
  - Reduce complexity of high-complexity algorithms

- **Outline**
  - Divide large problems into one or more subproblems of roughly the same size
    - E.g., split array into 2 halves, 3 thirds, etc.
  - Solve subproblems via recursion
  - Combine solutions to subproblems into solution for full problem
  - Solve small problems directly (*base case*)

- **Intuition**
  - If combining solutions is easier than solving directly, divide-and-conquer solution may be faster

Divide-and-conquer examples

- General strategy
  - Divide problem into equal parts
  - Solve subproblems recursively
  - Combine solutions
  - Solve small instances directly

- MergeSort
  - Split array in half
  - Recursively sort each half
  - Merge sorted arrays
  - Base case: 0 or 1 element

$2T(n/2)$

$\Theta(n)$

$T(n) = 2T(n/2) + \Theta(n)$

$T(n) = \Theta(n \lg n)$

Searching

- **Input**
  - *data*: sorted array of length *n*
  - *t*: target value
- **Output:** index *i* such that *data[i]* = *t*, or -1 if $t \notin data$

				t			

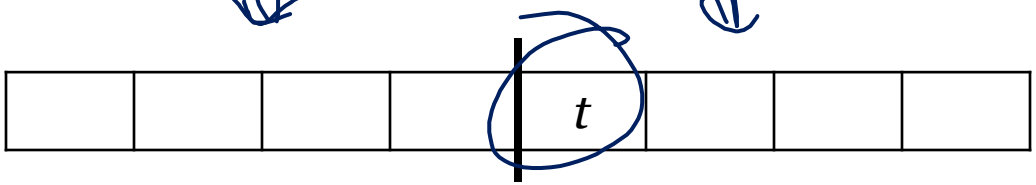
- Brute force
  - Check all values for *t*
  - Return when found
  - Worst-case: O(*n*)
- *Optimization:* stop searching if *data[i]* > *t*
  - Doesn't improve worst case

Input: $data$: sorted array to search
Input: n: length of $data$
Input: t: target value
Output: Index i such that $data[i] = t$, or -1 if $t \notin data$

```
Algorithm: ExhaustiveSearch
for i = 1 to n do
    if data[i] = t then
        return i
    end
end
return −1
```

Naïve binary search

- **Input**
  - *data*: sorted array of length *n*
  - *t*: target value
- **Output:** index *i* such that *data[i]* = *t*, or -1 if $t \notin data$



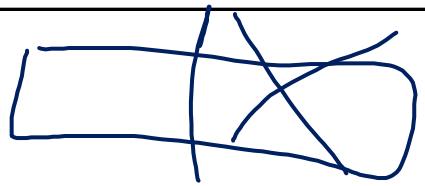
- Split *data* in half
- Search each half
- Return *t* if found
- Base case: array size 1

- $T(n) = 2T(n/2) + \Theta(1)$
  - $T(n) = \Theta(n)$ by Master Theorem
  - No better than linear search!

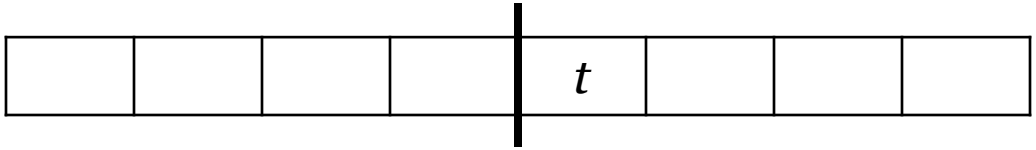
Input: $data$: sorted array to search
Input: n: length of $data$
Input: t: target value
Output: Index i such that $data[i] = t$, or -1 if $t \notin data$

```
Algorithm: NaïveBinSearch
if n = 1 then
    if data[1] = t then
        return 1
    else
        return −1
    end
end
mid = ⌊n/2⌋
lhs = NaïveBinSearch(data[1..mid])
rhs = NaïveBinSearch(data[mid + 1..n])
if lhs ≠ −1 then
    return lhs
else if rhs ≠ −1 then
    return mid + rhs
else
    return −1
end
```

Better binary search



- **Input**
  - *data*: sorted array of length n
  - t: target value
- **Output:** index i such that $data[i] = t$, or -1 if $t \notin data$



- *Observation*
  - In sorted array, only one half has t

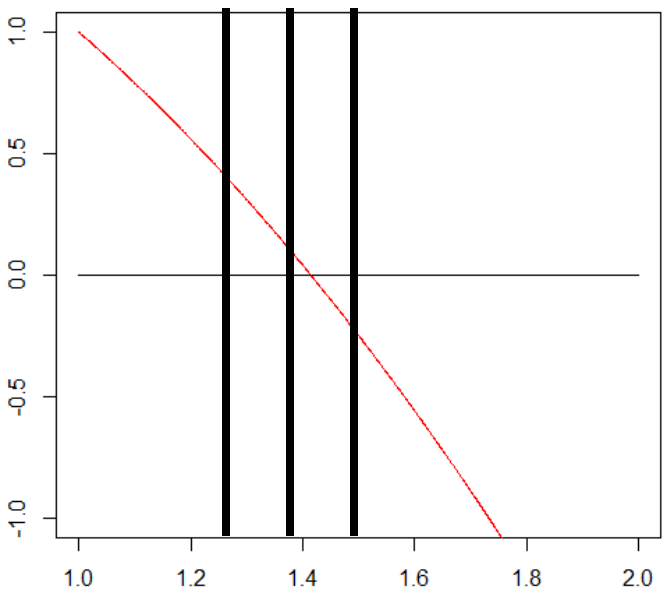
- Reduces number of recursive calls to 1
  - $T(n) = T(n/2) + \Theta(1)$
  - $T(n) = \Theta(\lg n)$ by MT

Input: *data*: sorted array to search
Input: n: length of *data*
Input: t: target value
Output: Index i such that $data[i] = t$, or -1 if $t \notin data$

```
Algorithm: BinSearch
if n = 1 then
    if data[1] = t then
        return 1
    else
        return −1
    end
end
mid = ⌊n/2⌋
if data[mid] = t then
    return mid
else if data[mid] > t then
    return BinSearch(data[1..mid − 1])
else
    return mid + BinSearch(data[mid + 1..n])
end
```

One-sided binary search

- Binary search can be applied to many different types of problems
  - E.g., estimating a root of $2-n^2$ between 1 and 2
    - Guess 1.5 (too high)
    - Guess 1.25 (too low)
    - Guess 1.375 (too low)
    - …
    - Repeat until sufficiently close
- Needs upper and lower bounds



- Alternative: one-sided search $f(1.25) = 0.4375$ $f(1.5) = -0.25$
  - Start with min value
  - Double until too large
  - Binary search between last success and first failure

One-sided binary search example

- Compute $\lfloor \lg n \rfloor$ for positive n
  - Lower bound?
  - Upper bound?

- Exhaustive solution
  - Try every power of 2 until you exceed n
  - $\Theta(\lg n)$ time

- One-sided binary search
  - *Double* power of 2 until too high
  - Binary search between last success and first failure
  - $\Theta(\lg \lg n)$ time

```
Input: n: positive integer
Output: ⌊lg n⌋
Algorithm: LinearScanLog
f = 1
x = 0
while f < n do
    f = 2f
    x = x + 1
end
return x
```

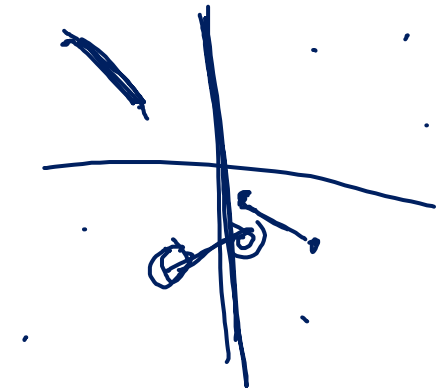
```
Input: n: positive integer
Output: ⌊lg n⌋
Algorithm: OneSidedLog
lo = 0
hi = 1
while 2^hi < n do
    lo = hi
    hi = 2hi
end
while hi − lo > 1 do
    mid = ⌊(hi + lo)/2⌋
    if 2^mid < n then
        lo = mid
    else
        hi = mid
    end
end
return lo
```

Divide-and-conquer analysis

- **Pros**
  - Reduces complexity for several problems
  - Easy to prove correctness via strong induction
  - Easy to analyze with Master Theorem
  - Works very well with parallel computing

$\Theta(n^2)$

n points



- **Cons**
  - Problem must exhibit *optimal substructure*
    - Solution must be related to subproblem solutions
  - Sometimes has poor complexity even with optimal substructure
    - *Dynamic programming* solves this problem

$\frac{n}{2} \cdot \frac{n}{2} = \frac{n^2}{4}$

$\rightarrow \Theta(n^2)$

Divide-and-conquer exercise

- Develop an efficient algorithm for *natural number exponentiation*
- **Input**
  - a: base of exponent (real number)
  - n: exponent (nonnegative integer)
- **Output:** a^n

- Divide problem into equal parts
- Solve subproblems recursively
- Combine solutions
- Solve small instances directly

- *Hint:* $a^n = \underbrace{a \cdot a \cdot a \cdots a}_{n \text{ times}}$
  - Focus on even n first
  - Think about how to handle odd n

```
Pow(a, n):
  if n = 0:
    return 1
  mid = ⌊n/2⌋
  t = Pow(a, mid)
  if n is even:
    return t^2
  else
    return a t^2
```

Divide-and-conquer exercise

- Develop an efficient algorithm for *natural number exponentiation*
- **Input**
  - a: base of exponent (real number)
  - n: exponent (nonnegative integer)
- **Output:** a^n

```
Input: a: base of exponent (real number)
Input: n: exponent (nonnegative integer)
Output: a^n
Algorithm: QuickPow
if n = 0 then
    return 1
end
t = QuickPow(a, ⌊n/2⌋)
if n is even then
    return t^2
else
    return at^2
end
```

Divide-and-conquer application

- Consider matrix addition and multiplication:
  - Given n-by-n matrices A and B, compute the sum $C = A + B$
  - Given an n-by-n matrices A and B, compute the product $C = AB$
- Addition
  - Compute $c_{ij} = a_{ij} + b_{ij}$, for every cij
  - Complexity: $\Theta(1)\Theta(n^2) = \Theta(n^2)$
- Multiplication
  - Compute c_{ij} = sum-product of row i of A and column j of B
    - $c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$
  - Complexity: $\Theta(n)\Theta(n^2) = \Theta(n^3)$
  - Fastest known algorithm until 1969

Strassen's Algorithm

- Described by Volker Strassen (1969)
- Used divide-and-conquer to reduce complexity of matrix multiplication
- Divide matrices into 4 quadrants:

$$\begin{array}{|c|c|}\hline A_{11} & A_{12} \\ \hline A_{21} & A_{22} \\ \hline \end{array} \times \begin{array}{|c|c|}\hline B_{11} & B_{12} \\ \hline B_{21} & B_{22} \\ \hline \end{array} = \begin{array}{|c|c|}\hline C_{11} & C_{12} \\ \hline C_{21} & C_{22} \\ \hline \end{array}$$

- Compute the following:

$$M_1 = (A_{11} + A_{22})(B_{11} + B_{22})$$
$$M_2 = (A_{21} + A_{22})B_{11}$$
$$M_3 = A_{11}(B_{12} - B_{22})$$
$$M_4 = A_{22}(B_{21} - B_{11})$$
$$M_5 = (A_{11} + A_{12})B_{22}$$
$$M_6 = (A_{21} - A_{11})(B_{11} + B_{12})$$
$$M_7 = (A_{12} - A_{22})(B_{21} + B_{22})$$

- Then:

$$C_{11} = M_1 + M_4 - M_5 + M_7$$
$$C_{12} = M_3 + M_5$$
$$C_{21} = M_2 + M_4$$
$$C_{22} = M_1 - M_2 + M_3 + M_6$$

Strassen's complexity

- Strassen's equations:

$$M_1 = (A_{11} + A_{22})(B_{11} + B_{22})$$
$$M_2 = (A_{21} + A_{22})B_{11}$$
$$M_3 = A_{11}(B_{12} - B_{22})$$
$$M_4 = A_{22}(B_{21} - B_{11})$$
$$M_5 = (A_{11} + A_{12})B_{22}$$
$$M_6 = (A_{21} - A_{11})(B_{11} + B_{12})$$
$$M_7 = (A_{12} - A_{22})(B_{21} + B_{22})$$

$$C_{11} = M_1 + M_4 - M_5 + M_7$$
$$C_{12} = M_3 + M_5$$
$$C_{21} = M_2 + M_4$$
$$C_{22} = M_1 - M_2 + M_3 + M_6$$

  - Total complexity: $T(n) = 7T(n/2) + \Theta(n^2)$
    - 18 additions/subtractions $\Theta(n^2)$
    - 7 multiplications $7T(n/2)$
- $c = \log_2(7) \approx 2.81$
- $f(n)$ vs. $n^{2.81} \Rightarrow f(n) = O(n^{\lg 7 - 0.8})$
- $T(n) = \Theta(n^c) \approx \Theta(n^{2.81})$

Dynamic programming

- Algorithm design strategy
- Similar to divide-and-conquer
- Useful when divide-and-conquer would be inefficient

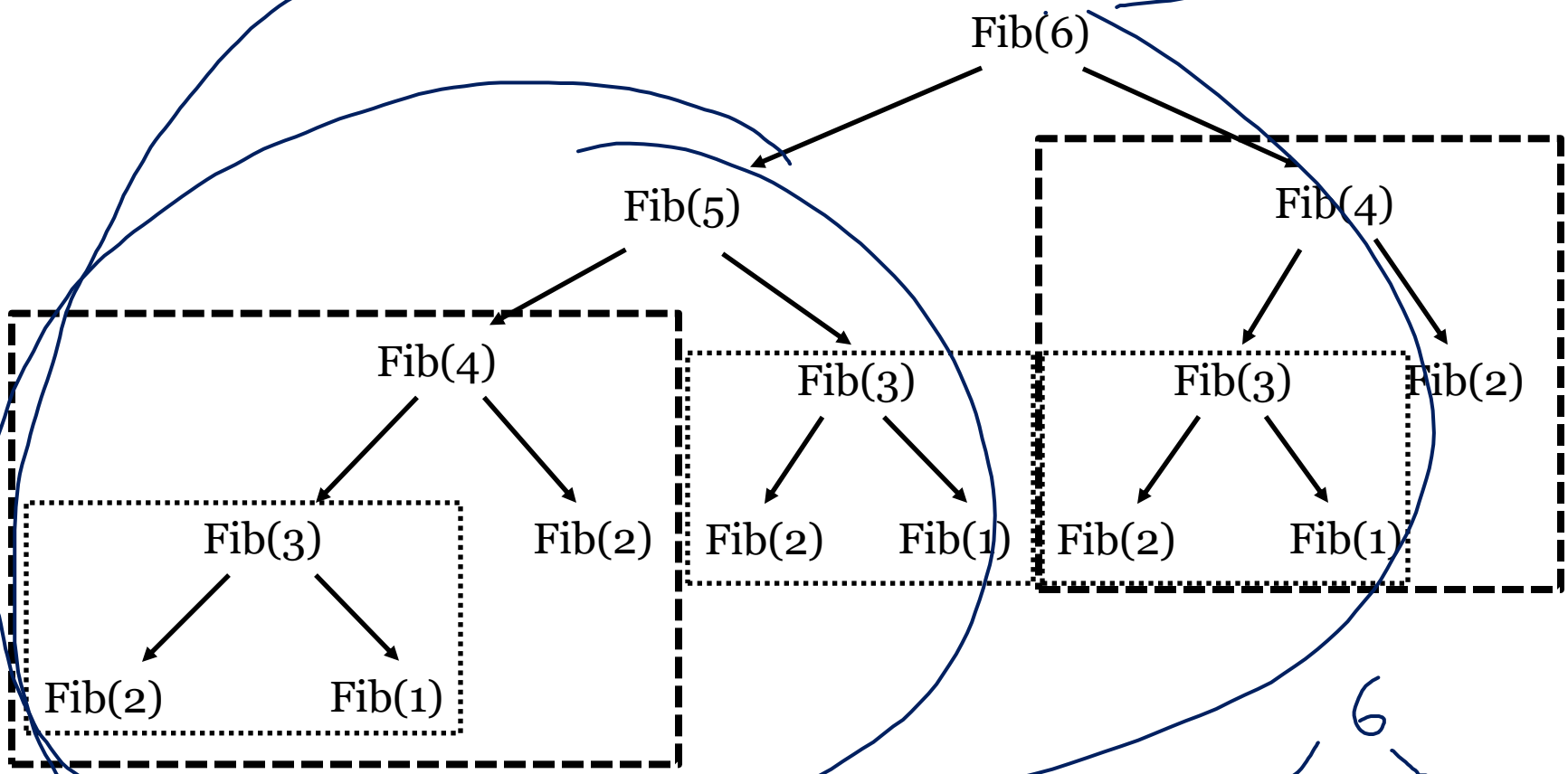
Dynamic programming motivation

$$F(n) = \begin{cases} 1, & \text{if } n = 1, 2 \\ F(n-1) + F(n-2), & \text{if } n > 2 \end{cases}$$

```
Input: n: index of Fibonacci number to compute
Output: F_n
Algorithm: Fibonacci
if n = 1 or 2 then
    return 1
else
    return Fibonacci(n − 1) + Fibonacci(n − 2)
end
```

- You've probably been told that this function is inefficient
- **Review:** recursive complexity
- *$T(n) = T(n-1) + T(n-2) + \Theta(1)$*
- Complexity: $\Theta(\tau^n)$, where $\tau = \frac{1+\sqrt{5}}{2} \approx 1.61$
  - Calculating F_{1000} takes forever!

Why does this take so long?



- We're computing the same sub-results over and over!
- **Intuition:** save sub-results in a map

A better solution

~~-1~~	-1	~~-1~~	-1	-1	-1	-1
1	2	3	4	5	6	7
1	1	2	3	5	8	13

Input: n: index of Fibonacci number to compute
Output: F_n

Algorithm: FastFib
$fib = \text{Array}(n)$
Initialize fib to -1
return DynamicFib(n)

Algorithm: DynamicFib(n)
if $fib[n] = -1$ **then**
  if $n = 1$ or 2 **then**
    $fib[n] = 1$
  else
    $fib[n] = \text{DynamicFib}(n-1) + \text{DynamicFib}(n-2)$
  end
end
return $fib[n]$

A better solution

1	1	2	3	5	8	13
1	2	3	4	5	6	7

Input: n: index of Fibonacci number to compute
Output: F_n

```
Algorithm: FastFib
fib = Array(n)
Initialize fib to −1
return DynamicFib(n)
```

```
Algorithm: DynamicFib(n)
if fib[n] = −1 then
    if n = 1 or 2 then
        fib[n] = 1
    else
        fib[n] = DynamicFib(n − 1) +
          DynamicFib(n − 2)
    end
end
return fib[n]
```

- Each Fibonacci number computed *once*
- Total complexity: $\Theta(n)$

Dynamic programming

- General algorithm design strategy
- Recursion augmented with data structure to save sub-results
  - Not always a 1D array…
    - Depends on number of parameters
- **Useful when:**
  - Problem has recursive structure (*like divide-and-conquer*)
  - More than one recursive call
  - Problems overlap
    - Typically, 1-2 levels deep in recursion tree
- **Related concept:** memoization
  - Function that stores sub-results
  - Returns immediately when calling with previously-seen value
  - *Note:* possible to memoize non-recursive functions
    - Dynamic programming can be implemented iteratively

A more complex example

- Binomial coefficients ("n choose k")
  - **Formula:** $\binom{n}{k} = \frac{n!}{k!(n-k)!}$
  - **Example:** $\binom{20}{3} = \frac{20!}{3!(17!)}$

$$= \frac{2432902008176640000}{6(355687428096000)}$$

$$= 1140$$

- Formula might cause overflow
- Alternative: Pascal's triangle

k 0 1 2 3

n 0 1 2 3

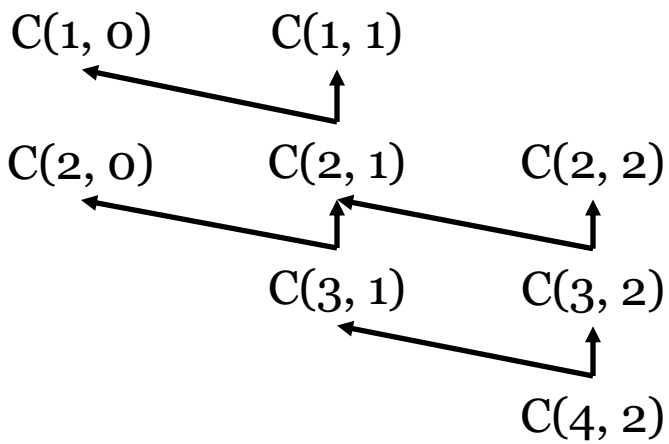
			1			
		1		1		
	1		2		1	
1		3		3		1
1	4		6		4	1

$$\binom{n}{k} = \begin{cases} 1, \text{ if } k = 0 \text{ or } k = n \\ \binom{n-1}{k-1} + \binom{n-1}{k} \end{cases}$$

Binomial coefficients example

- Recursive function
- **Example:** C(4, 2)

$$\binom{n}{k} = \begin{cases} 1, \text{ if } k = 0 \text{ or } k = n \\ \binom{n-1}{k-1} + \binom{n-1}{k} \end{cases}$$



- Overlapping subproblems
  - Dynamic programming!

Binomial coefficients example

- Recursive function
- **Example:** C(4, 2)

$$\binom{n}{k} = \begin{cases} 1, \text{ if } k = 0 \text{ or } k = n \\ \binom{n-1}{k-1} + \binom{n-1}{k} \end{cases}$$

	0	1	2
0			
1	C(1, 0)	C(1, 1)	
2	C(2, 0)	C(2, 1)	C(2, 2)
3		C(3, 1)	C(3, 2)
4			C(4, 2)

- Overlapping subproblems
  - Dynamic programming!
- Data structure?
  - 2D array (map)

Binomial coefficients example

Recurrence: $$\binom{n}{k} = \begin{cases} 1, \text{ if } k = 0 \text{ or } k = n \\ \binom{n-1}{k-1} + \binom{n-1}{k} \end{cases}$$

```
Algorithm: BasicBinom
if k = 0 or k = n then
    return 1
else
    return BasicBinom(n − 1, k − 1) +
      BasicBinom(n − 1, k)
end
```

- Using 2D array:

```
Algorithm: AlmostMemo
if we've already calculated binom[n, k] then
    return binom[n, k]
else if k = 0 or k = n then
    binom[n, k] = 1
else
    binom[n, k] = AlmostMemo(n − 1, k − 1) +
      AlmostMemo(n − 1, k)
end
return binom[n, k]
```

```
Algorithm: MemoBinom
if binom[n, k] > 0 then
    return binom[n, k]
else if k = 0 or k = n then
    binom[n, k] = 1
else
    binom[n, k] = MemoBinom(n−1, k−1)
      + MemoBinom(n − 1, k)
end
return binom[n, k]
```

- Final answer

```
Input: n, k: binomial coefficient to compute
Output: (n k)
Algorithm: DPBinom
binom = Array(n, k)
Initialize binom to 0
return MemoBinom(n, k)
```

Memoization summary

- Five step procedure

1. **Start with naïve recursive algorithm**
   - Based on recurrence
2. **Decide data structure and sentinel value**
3. **Add "memoization check" to the beginning of the algorithm**
   - If solution is in data structure, return it
4. **Store solution before returning**
5. **Write wrapper function to initialize data structure**

$$\binom{n}{k} = \begin{cases} 1, \text{ if } k = 0 \text{ or } k = n \\ \binom{n-1}{k-1} + \binom{n-1}{k} \end{cases}$$

Input: n, k: binomial coefficient to compute
Output: $\binom{n}{k}$

Algorithm: DPBinom
$binom = \text{Array}(n, k)$
Initialize $binom$ to 0
return MemoBinom(n, k)

Algorithm: MemoBinom
if $binom[n, k] > 0$ **then**
  return $binom[n, k]$
else if $k = 0$ or $k = n$ **then**
  $binom[n, k] = 1$
else
  $binom[n, k] = \text{MemoBinom}(n-1, k-1)$
  $+ \text{MemoBinom}(n-1, k)$
end
return $binom[n, k]$

Memoization exercise

- Stirling Numbers of the Second Kind, *S(n, k),* count the number of ways to divide *n* objects into *k* groups.
- They satisfy the following recurrence:

$$S(n,k) = \begin{cases} 1, & \text{if } k = 1 \text{ or } k = n \\ kS(n-1,k) + S(n-1,k-1), & \text{otherwise} \end{cases}$$

1. Give pseudocode for a naïve recursive algorithm that computes *S(n, k)* using this recurrence.
2. What map implementation would you use for a memoized dynamic programming solution to this problem and why?
3. What would be a reasonable sentinel value to represent an unsolved problem in this map?
4. Give pseudocode for a memoized dynamic programming algorithm for this problem.

Memoization exercise

$$S(n,k) = \begin{cases} 1, & \text{if } k = 1 \text{ or } k = n \\ kS(n-1,k) + S(n-1,k-1), & \text{otherwise} \end{cases}$$

1. Naïve pseudocode
2. Map implementation
3. Sentinel value
4. Memoized pseudocode

```
Stirling(n, k):
memo = Array(n, k)
Initialize memo to 0
return S(n, k, memo)

S(n, k, memo):
if memo[n, k] != 0:
  return memo[n, k]
if k = 1 or k = n:
  memo[n, k] = 1
else:
```

Memoization exercise

1.

```
Input: n: number of objects (positive integer)
Input: k: number of groups (positive integer)
Output: S(n, k)
Algorithm: Stirling
if k = 1 or k = n then
    return 1
else
    return k · Stirling(n − 1, k) + Stirling(n − 1, k − 1)
end
```

2. 2D array-based map (n by k) because both n and k vary

3. -1 (# of ways to divide things can't be negative); 0 or NIL work, too

4.

```
Input: n: number of objects
       (positive integer)
Input: k: number of groups
       (positive integer)
Output: S(n, k)
Algorithm: Stirling2
S = Array(n, k)
Initialize S to −1
return MemoStirling(n, k)
```

```
Input: n: number of objects (positive integer)
Input: k: number of groups (positive integer)
Output: S(n, k)
Algorithm: MemoStirling
if S[n, k] = −1 then
    if k = 1 or k = n then
        S[n, k] = 1
    else
        S[n, k] = k · MemoStirling(n − 1, k) +
          MemoStirling(n − 1, k − 1)
    end
end
return S[n, k]
```

Dynamic programming example

- Longest common subsequence
  - Given two strings, *a* and *b*
  - Compute string *c* so that characters of *c* appear in both
    - In same order
- **Example**

g	a	r	b	a	g	e

g	a	r				e

		g	a	r		e	

b	o	g	a	r	t	e	d

- **LCS:** gare
- **Main idea:** consider each decision independently
  - Match if possible
  - If not, skip a_i or b_j
  - Find LCS for remainder

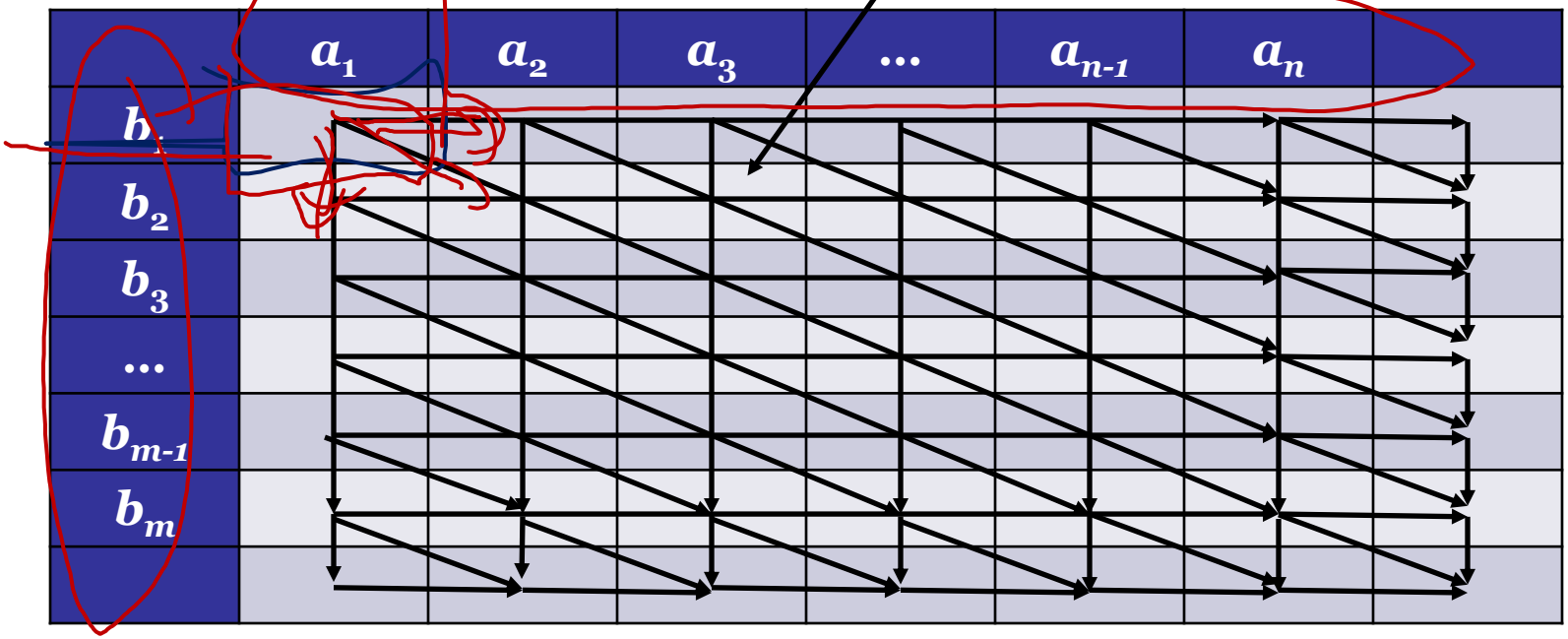
Dependency structure

- $LCS(a,b) = \max \begin{cases} a_1 + LCS(a',b'), & \text{if } a_1 = b_1 \\ LCS(a',b) \\ LCS(a,b') \end{cases}$
- Base cases: $LCS(a,\epsilon) = \epsilon$

  $LCS(\epsilon,b) = \epsilon$

Two arguments:
2D structure

Subproblems:
find LCS of suffix



Tracing example

- LCS(air, care):

	a	*i*	*r*	
c				
a				
r				
e				

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c				ϵ
a				
r				
e				

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c				ϵ
a				ϵ
r				
e				

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c				ϵ
a				ϵ
r				
e				ϵ

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c				ϵ
a				ϵ
r			*r*	
e				ϵ

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c				ϵ
a			*r*	ϵ
r			↑ *r*	
e				ϵ

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c			*r*	ϵ
a			↑ *r*	ϵ
r			*r*	
e				ϵ

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c			*r*	ϵ
a			*r*	ϵ
r			*r*	
e				ϵ
			ϵ	

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c			*r*	ϵ
a			*r*	ϵ
r			*r*	
e			ϵ	ϵ
			ϵ	

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c			*r*	ϵ
a			*r*	ϵ
r			*r*	
e			ϵ	ϵ
		ϵ	ϵ	

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c			*r*	ϵ
a			*r*	ϵ
r			*r*	
e		ϵ	ϵ	ϵ
		ϵ	ϵ	

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c			*r*	ϵ
a			*r*	ϵ
r		*r*	*r*	
e		ϵ	ϵ	ϵ
		ϵ	ϵ	

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c			*r*	ϵ
a		*r*	*r*	ϵ
r		*r*	*r*	
e		ϵ	ϵ	ϵ
		ϵ	ϵ	

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c		*r*	*r*	ϵ
a		↑ *r*	*r*	ϵ
r		*r*	*r*	
e		ϵ	ϵ	ϵ
		ϵ	ϵ	

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c		*r*	*r*	ϵ
a	*ar*	*r*	*r*	ϵ
r		*r*	*r*	
e		ϵ	ϵ	ϵ
		ϵ	ϵ	

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c	*ar*	*r*	*r*	ϵ
a	*ar*	*r*	*r*	ϵ
r		*r*	*r*	
e		ϵ	ϵ	ϵ
		ϵ	ϵ	

Tracing example

- LCS(air, care):

	a	*i*	*r*	
c	*ar*	*r*	*r*	ϵ
a	*ar*	*r*	*r*	ϵ
r		*r*	*r*	
e		ϵ	ϵ	ϵ
		ϵ	ϵ	

air
aardvark
car
carvark

LCS: memoization

- $LCS(a,b) = \max \begin{cases} a_1 + LCS(a',b'), & \text{if } a_1 = b_1 \\ LCS(a',b) \\ LCS(a,b') \end{cases}$

<u>Base cases</u>

$LCS(a,\epsilon) = \epsilon$

$LCS(\epsilon,b) = \epsilon$

- Implement recurrence directly
- Rely on matrix for previous subproblems
- Write outer function to initialize matrix
  - Need to choose sentinel value carefully

```
Algorithm: MemoLCS(a, i, b, j)
if lcs[i,j] ≠ NIL then
    return lcs[i,j]
else if i = n + 1 or j = m + 1 then
    lcs[i,j] = ϵ
else if a[i] = b[j] then
    lcs[i,j] = a[i] + MemoLCS(a, i + 1, b, j + 1)
else
    o1 = MemoLCS(a, i + 1, b, j)
    o2 = MemoLCS(a, i, b, j + 1)
    lcs[i,j] = max(o1, o2)
end
return lcs[i,j]
```

```
Input: a, b: two strings
Input: n, m: length of a and b,
       respectively
Output: LCS of a and b
Algorithm: DPLCS
lcs = Array(n, m)
Initialize all cells of lcs to NIL
return MemoLCS(a, 1, b, 1)
```

An alternative approach

- Consider the problem of computing the *size* of the LCS
- Solution is very similar:

$$LCSlen(i,j) = \begin{cases} 0, & \text{if } i = n+1 \text{ or } j = m+1 \\ 1 + LCSlen(i+1, j+1), & \text{if } a_i = b_j \\ \max(LCSlen(i+1,j), LCSlen(i,j+1)), & \text{otherwise} \end{cases}$$

	a	*i*	*r*	
c	*ar*	*r*	*r*	ε
a	*ar*	*r*	*r*	ε
r		*r*	*r*	
e		ε	ε	ε
		ε	ε	ε

→

	a	*i*	*r*	
c	2	1	1	0
a	2	1	1	0
r		1	1	
e		0	0	0
		0	0	0

+

	a	*i*	*r*	
c	↓	→	↓	
a	↘	→	↓	
r		→	↘	
e				

- **Observation:** we can reconstruct LCS string if we also store where max came from
  - Follow "pointers" from start to base case
  - Add characters when we move diagonally
- Extra $\Theta(n+m)$ processing step (doesn't change complexity)
- Reduces space complexity

Coming up

- Dynamic programming
  - Longest common subsequence
  - Iterative DP
- Greedy algorithms
- *Recommended readings:* Sections 11.3, 11.2, 12.1, 12.2, 12.5
  - *Practice problems:* R-11.3, C-11.3 (complexity only), C-11.4, A-11.2, R-12.9, C-12.6, C-12.7, C-12.9, C-12.10, A-12.1, A-12.2, A-12.5